

Peer-to-peer protocols for efficient and resilient decentralized learning

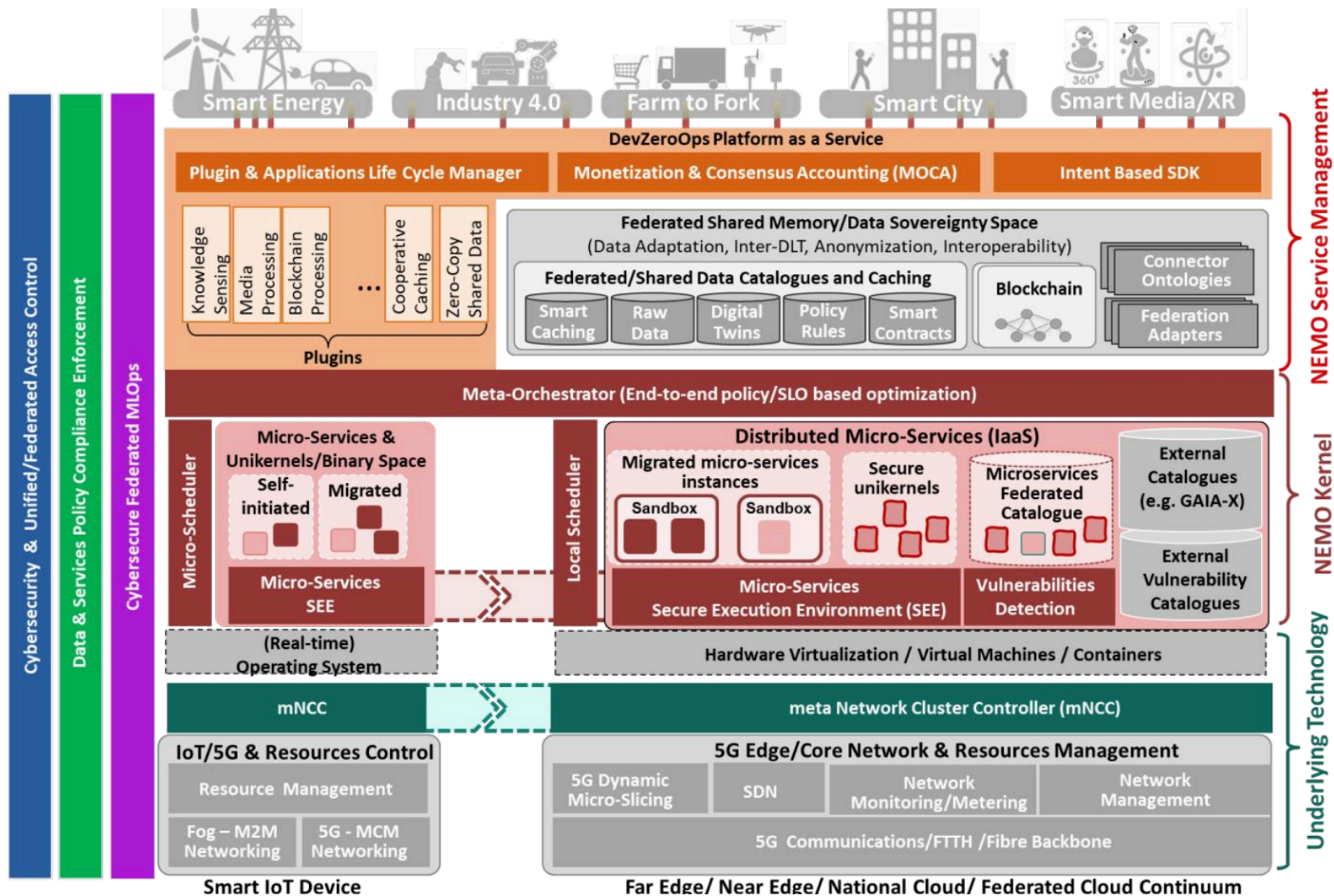
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LIP6 · Sorbonne Université

September 7, 2026

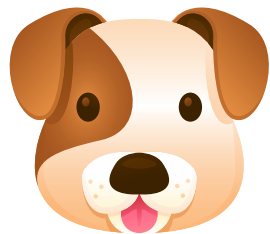
Context



Supervised Classification

Training

Data

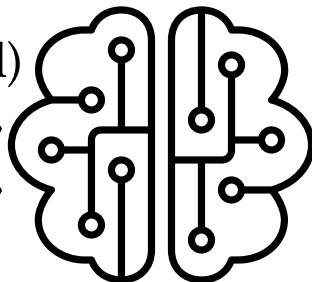


Dog



Cat

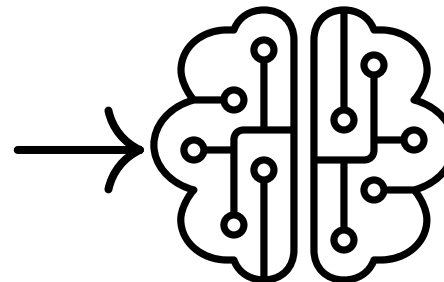
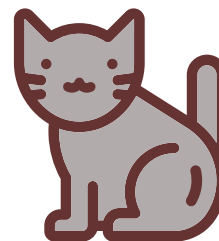
(data, label)



Model

Labels

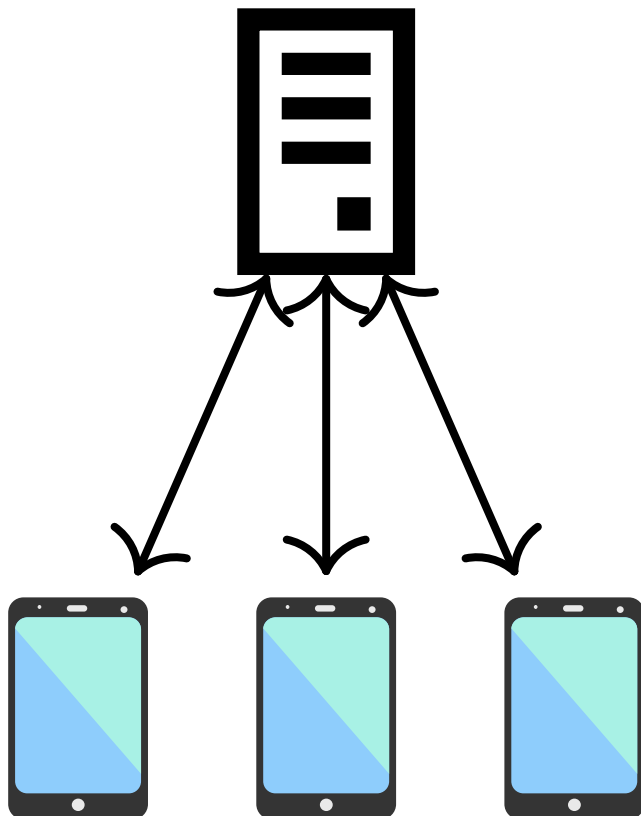
Prediction



Trained Model

Cat

Federated Learning



Server

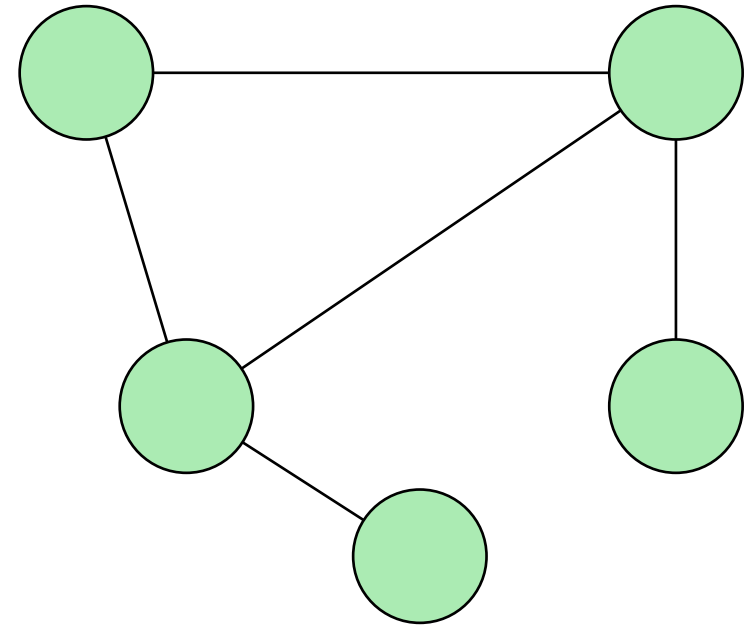
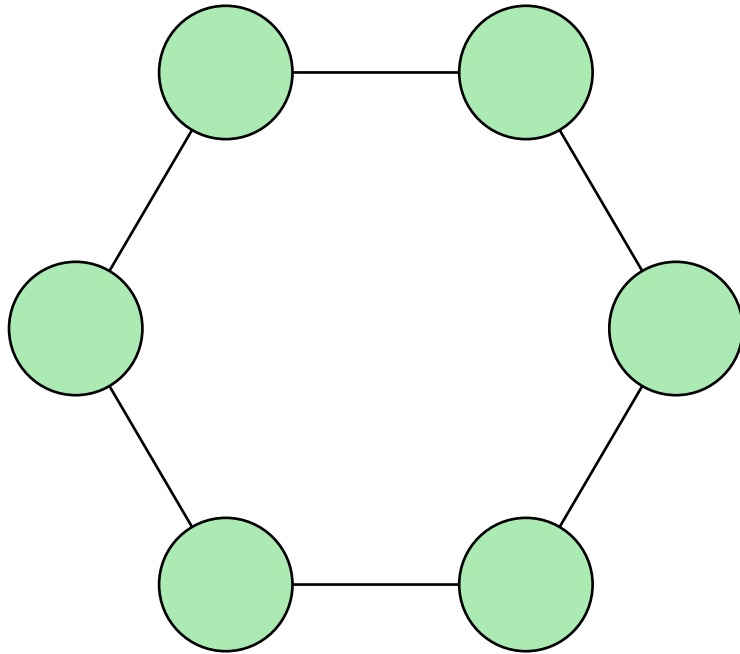
```
1 loop
2   received_models  $\leftarrow$  collect(clients)
3   global_model  $\leftarrow$  merge(received_models)
4   broadcast(global_model)
```

Client

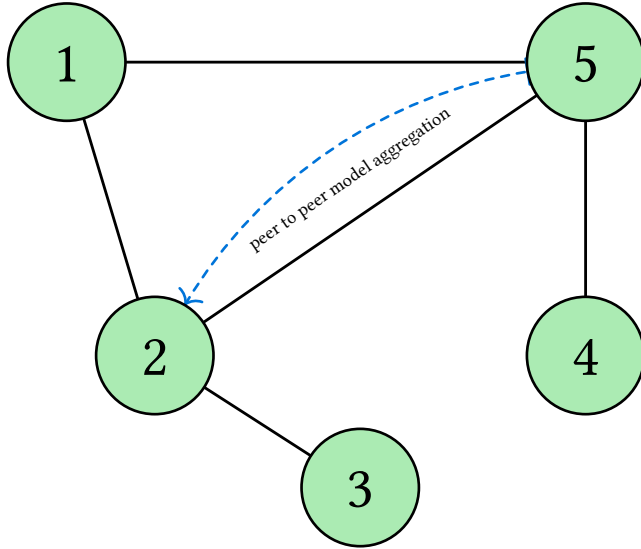
```
1 loop
2   local_model  $\leftarrow$  train(model, local_data)
3   send(local_model, server)
4   model  $\leftarrow$  receive(global_model)
```

[1] B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. y Arcas, “Communication-efficient learning of deep networks from decentralized data,” in *Artificial intelligence and statistics*, 2017, pp. 1273–1282.

Structured vs Unstructured networks



Gossip Learning



Gossip Learning (main thread)

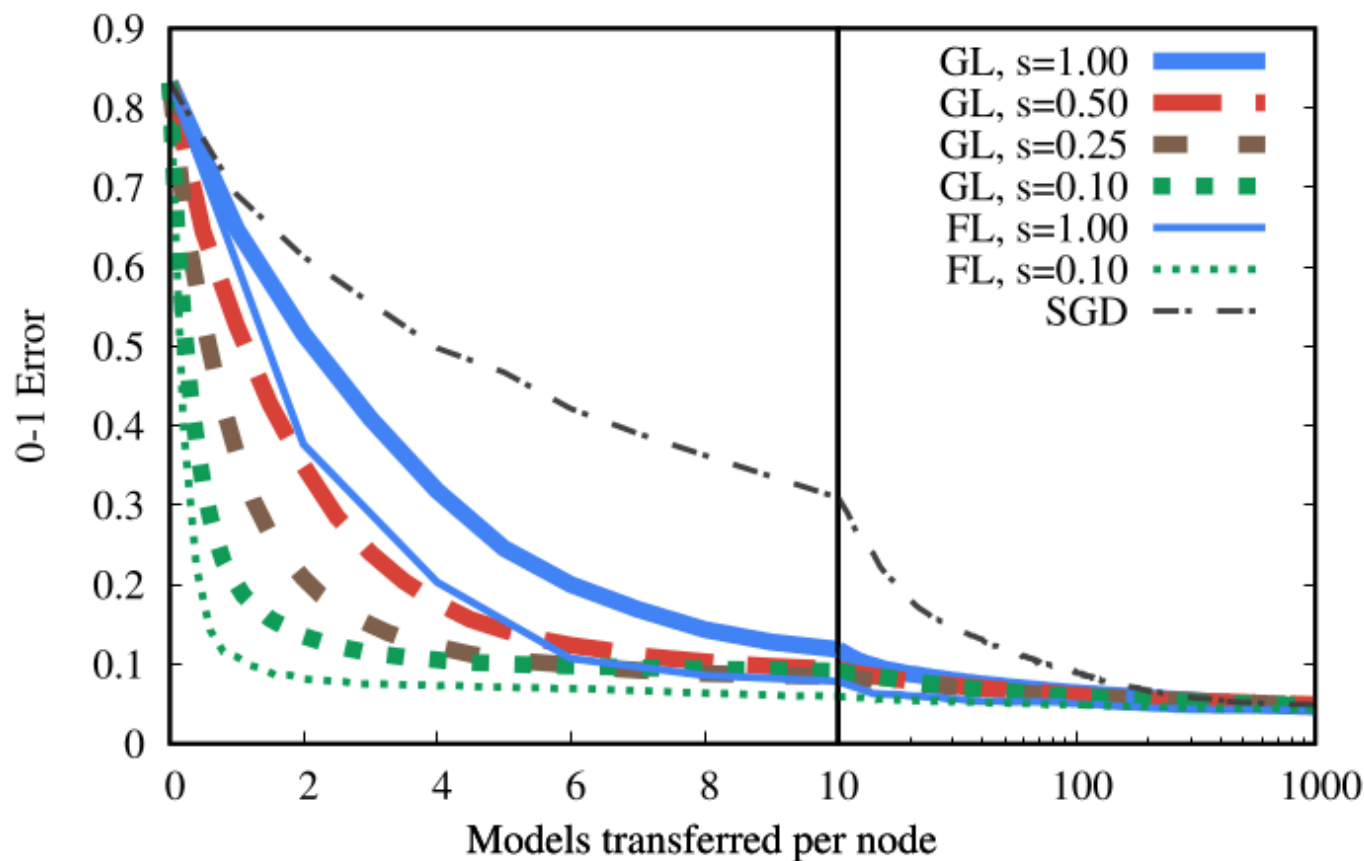
ML model: **model**

List of neighbors : **cache**

```
1 loop
2   | Wait for  $\Delta$  time units
3   |  $\text{peer} \leftarrow \text{selectRandom}(\text{cache})$ 
4   |  $\text{send}(\text{peer}, \text{model})$ 
```

[2] R. Ormándi, I. Hegedűs, and M. Jelasity, “Gossip learning with linear models on fully distributed data,” *Concurrency and Computation: Practice and Experience*, vol. 25, no. 4, pp. 556–571, 2013.

Gossip Learning vs Federated Learning

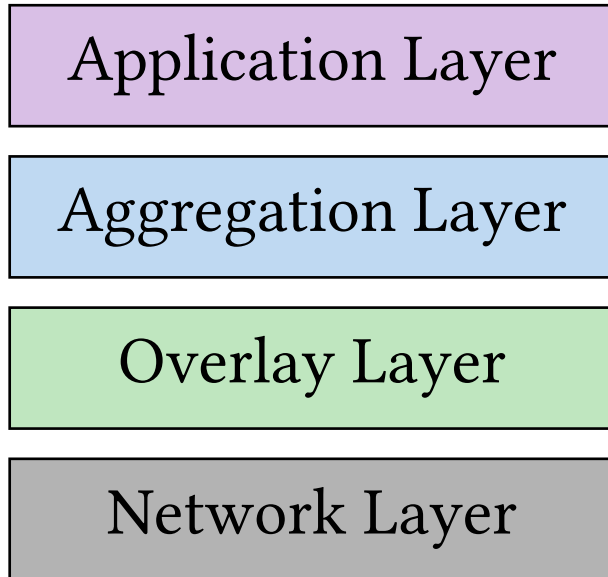


[3] I. Hegedűs, G. Danner, and M. Jelasity, “Decentralized learning works: An empirical comparison of gossip learning and federated learning,” *Journal of Parallel and Distributed Computing*, vol. 148, pp. 109–124, 2021.

State of the Art Summary

	Decentralized	Local data	Fast convergence	Fault-tolerant	No overlay overhead
Centralized learning	✗	✗	✓	✗	✓
Federated learning	✗	✓	✓	✗	✓
Decentralized learning (structured net.)	✓	✓	✓	✗	✗
Gossip learning	✓	✓	✗	✓	✓
Local learning	✓	✓	✗ no generalizing model	✓	✓

No single solution is fully satisfactory



Outline

- **Elevator** — overlay (hub election)
- **Lift** — malicious resilience
- **HEAL** — model aggregation
- **FLAIR** — wireless networks adaptation

Elevator

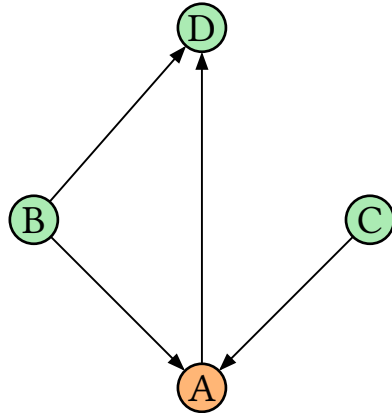
Definition. (Hub) Let $G = (V, E)$ be the directed overlay graph, where each node $v \in V$ keeps a partial view $C(v) \subseteq V$.

A node $h \in V$ is a **hub** if it appears in the partial view of every node:

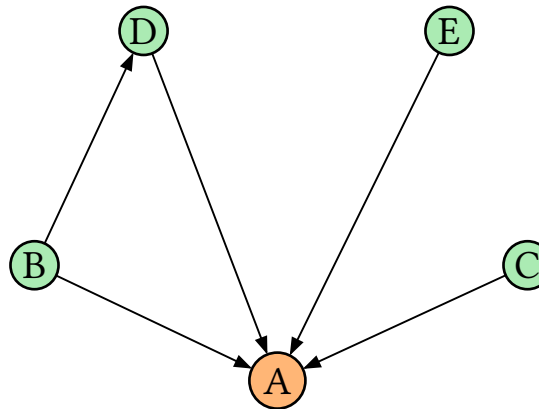
$$\forall v \in V, \quad h \in C(v)$$

[4] M. A. Legheraba, M. Potop-Butucaru, S. Tixeuil, and S. Fdida, “Emergent Peer-to-Peer Multi-Hub Topology,” in *2024 22nd International Symposium on Network Computing and Applications (NCA)*, 2024, pp. 219–226.

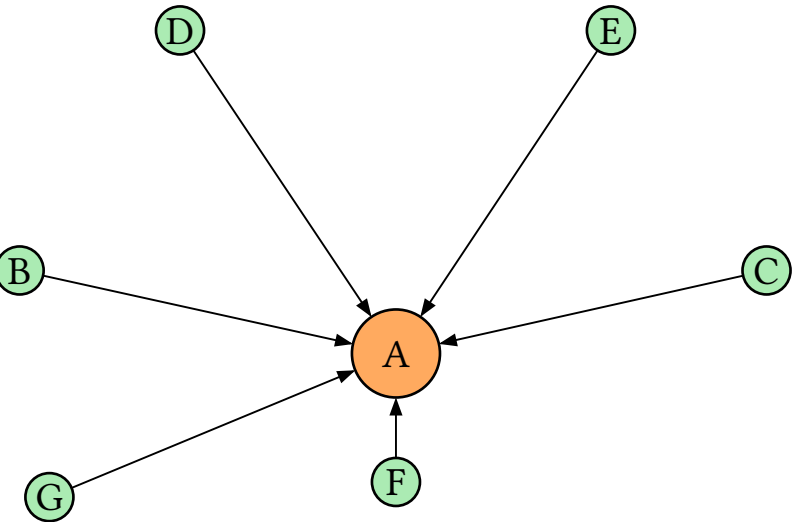
1. One node is more connected



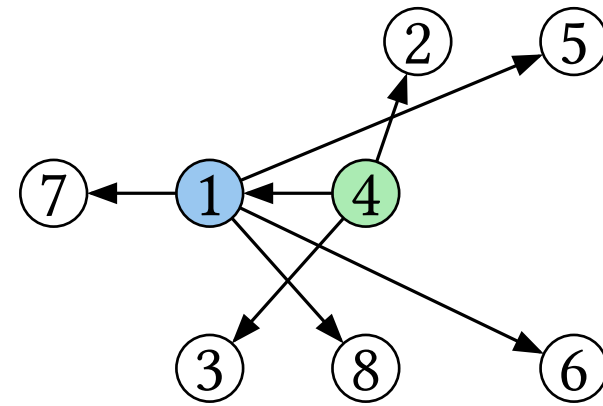
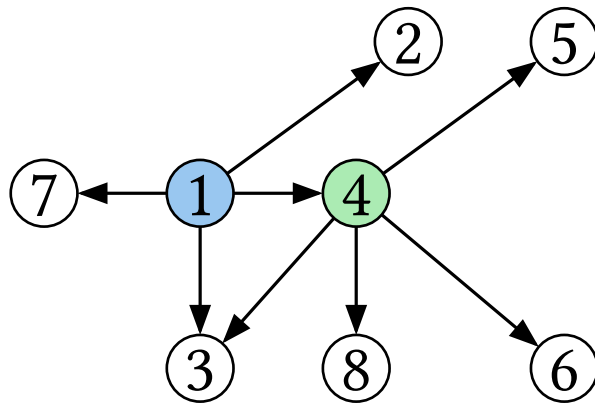
2. It attracts more connections



3. Snowball effect: it becomes a hub



[5] A.-L. Barabási, H. Jeong, Z. Néda, E. Ravasz, A. Schubert, and T. Vicsek, “Evolution of the social network of scientific collaborations,” *Physica A: Statistical mechanics and its applications*, vol. 311, no. 3–4, pp. 590–614, 2002.



Before and after a shuffling operation. Node 1 sends addresses {itself, 2, 3} to node 4. Node 4 sends back {5,6, 8}.

[6] A. Stavrou, D. Rubenstein, and S. Sahu, “A lightweight, robust p2p system to handle flash crowds,” in *10th IEEE International Conference on Network Protocols, 2002. Proceedings.*, 2002, pp. 226–235.

Preferential attachment makes hubs emerge...

Random attachment keeps the network resilient...

Can we combine both to get **hubs** that are
resilient?

Mechanism

1. Each node asks its **neighbours** for **their** neighbour lists.
2. It connects to the **h most frequent** nodes.
3. It asks these h hubs for extra incoming connections, to fill a **random subset** of its view.

Observed

- After a few cycles, **hubs emerge** (the preferred h are the same for everyone);
- Hubs are **elected at random** among the network nodes;
- Even after **hub failures**, new nodes are elected as hubs.



Example of topology generated

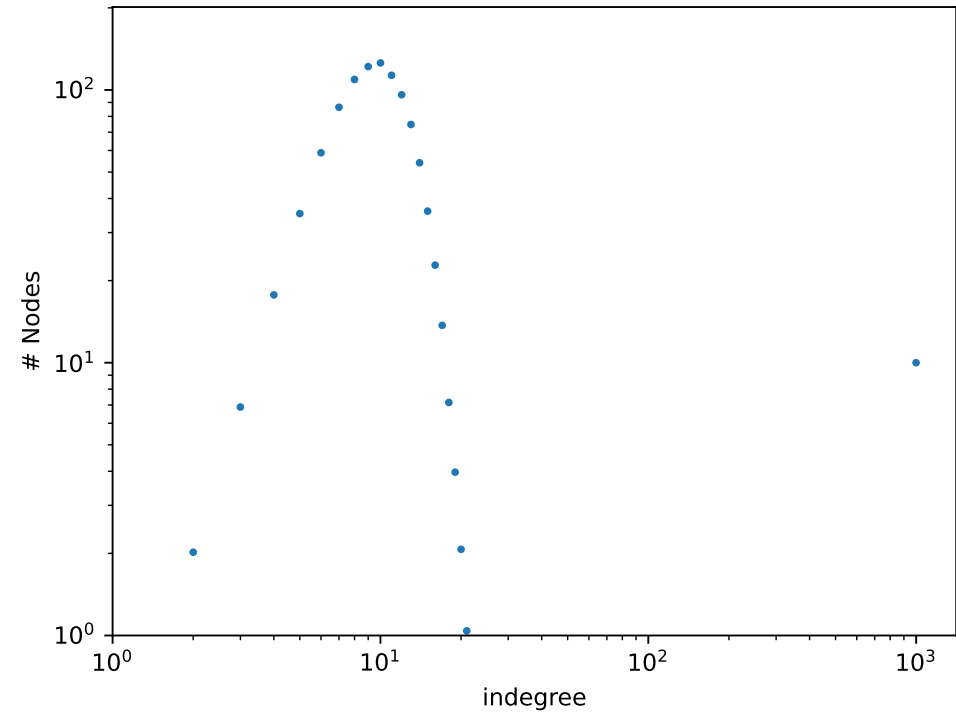
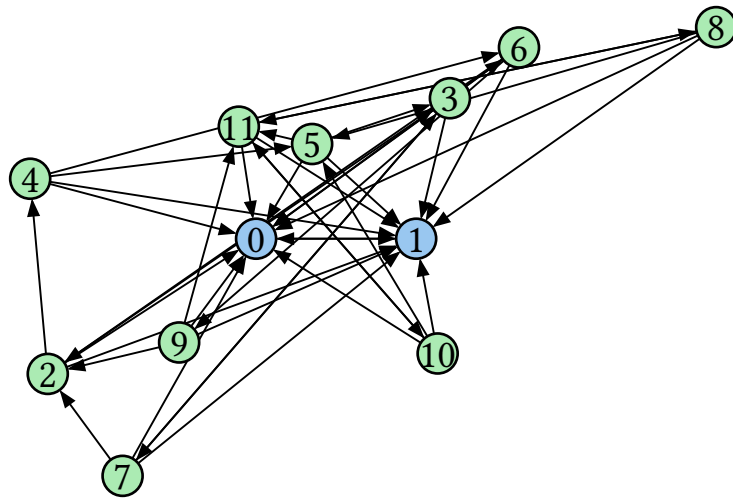


Figure 1: Indegree distribution of a network generated with Elevator, with 1000 nodes and 10 hubs

Definition. (Stability) Once Elevator has converged to a set of h hubs, both the list of hubs and their number h remain constant (with high probability) over time.

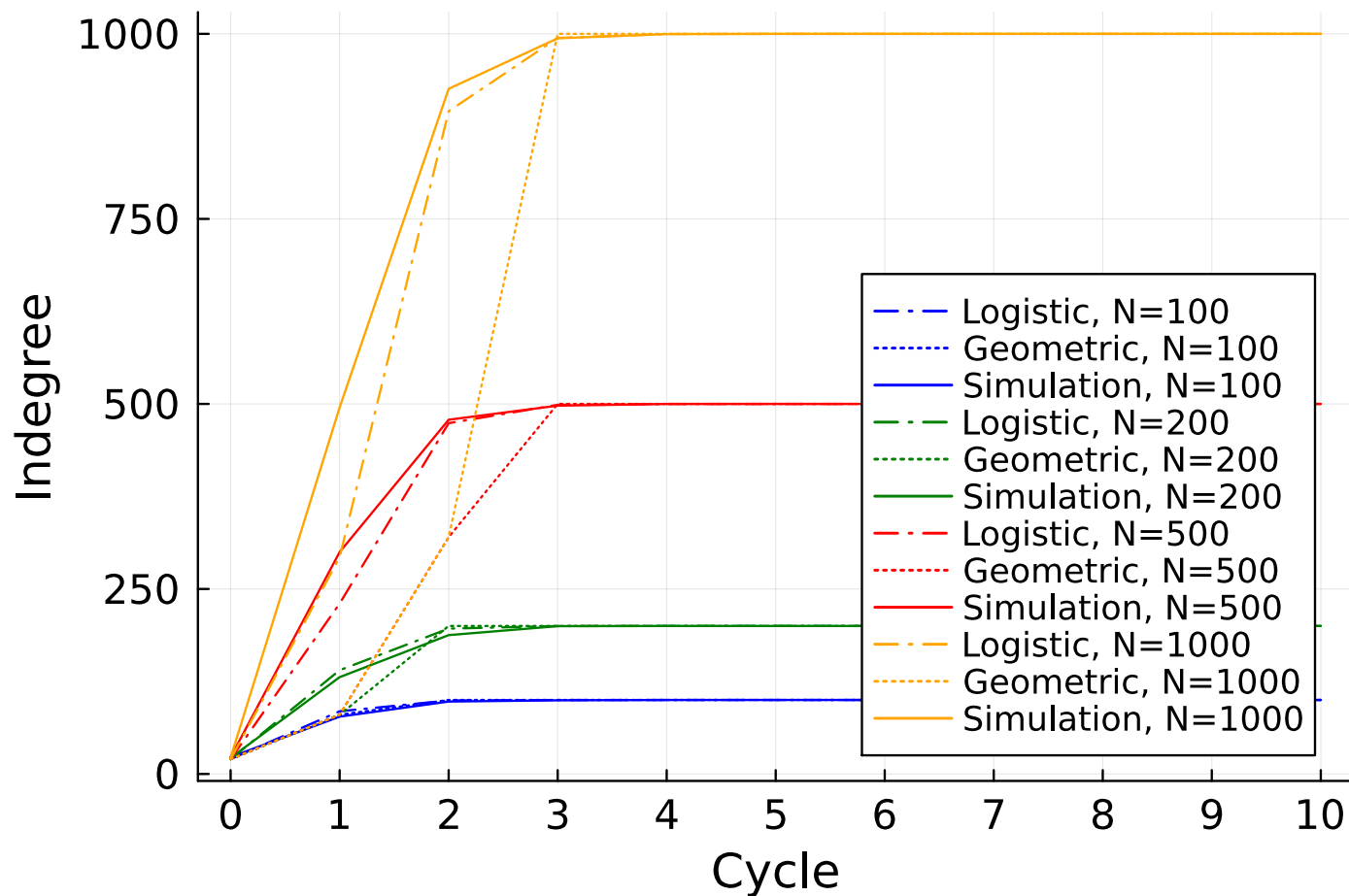
Formally, after convergence time T :

$$\Pr[\forall t \geq T, H(t) = H(T) \wedge \forall v \in V, C_v(t)[1 : h] = H(t)] = 1$$

Definition. (Convergence) The network converges to a stable state containing **exactly** h hubs shared by all nodes.

Idea of the proof.

- A new hub is produced with **non-zero probability** at each cycle.
 - The number of hubs **cannot decrease** (until it reaches h).
- Once at least one hub exists, the network is **strongly connected** (w.h.p.) and produces a new hub.



PeerSim was **forked and substantially rewritten** for this thesis:

- Migrated from **SVN to Git**, build rewritten in **Gradle**;
- **Docker + GitLab CI/CD**;
- **Parallelised** the cycle-based engine;
- Failure models from scratch (**crash, churn, Byzantine**);

Parameters

- Network size: **$N = 1000$ nodes**;
- **1000 cycles**, repeated **100 times**;
- Initial topology: **random k -out graph** with **$k = c = 20$** ;
- **$h = 10$ hubs**;
- e.g. brutal crash: **50% of nodes** disconnected at cycle **500**.

Overlay

- In-degree distribution
- Clustering coefficient
- Average path length
- Diameter
- Convergence time

Machine learning



- Accuracy
- Convergence time

Overlay

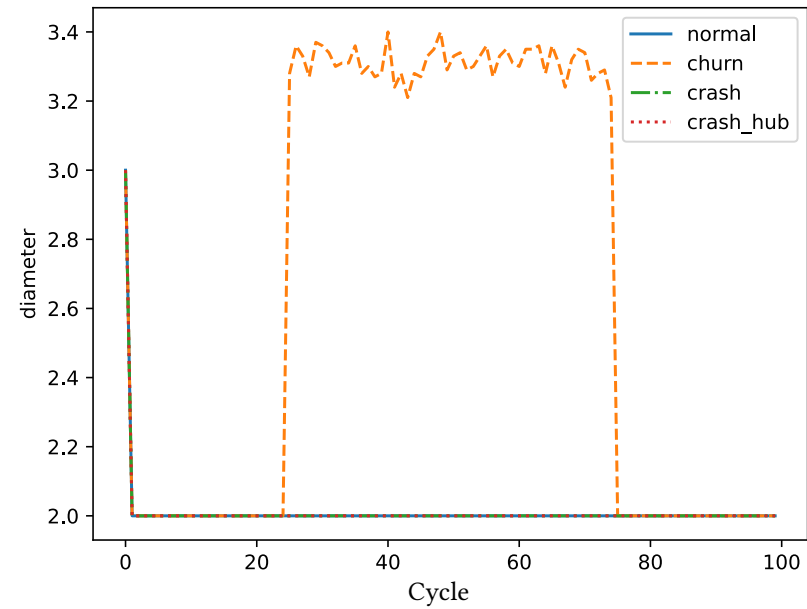
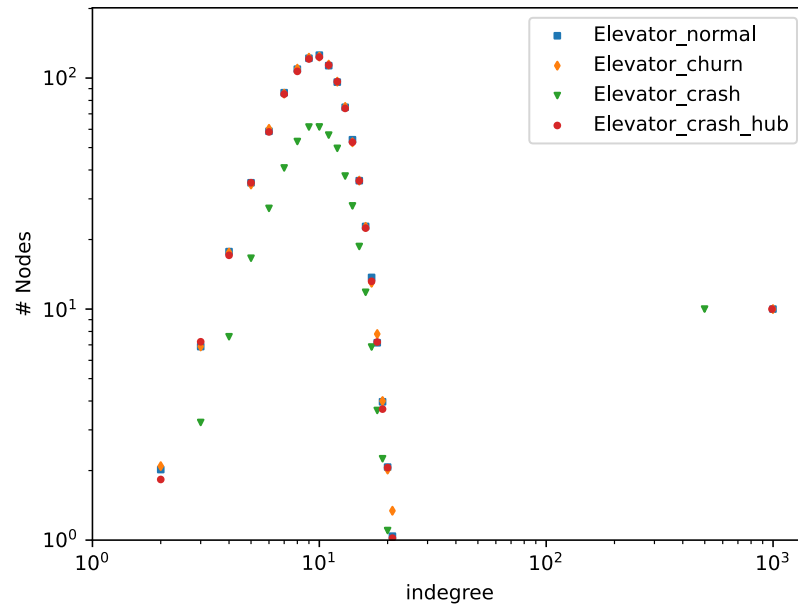
- **Crash failures**
- **Churn** (dynamic joins / leaves)
- **Byzantine failures**

Machine learning

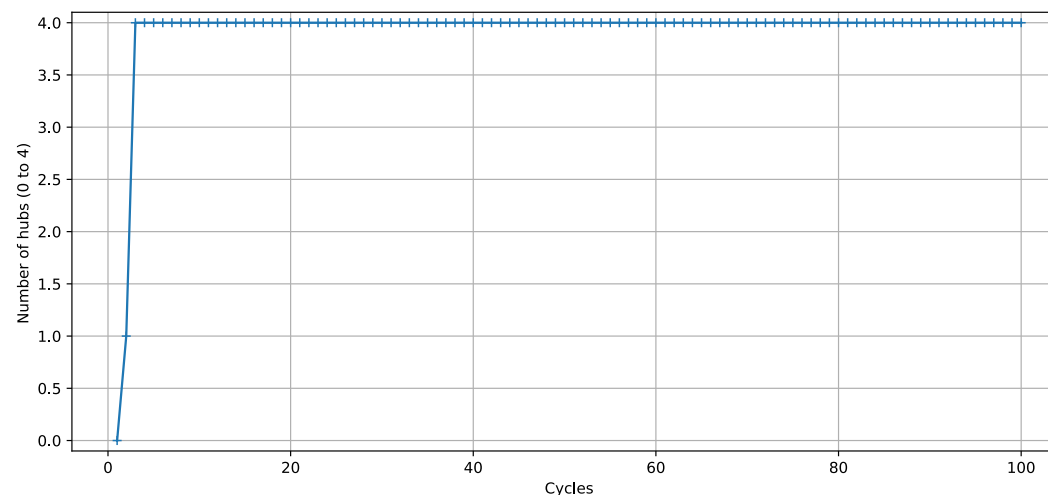


- **Privacy attacks**
- **Poisoning attacks**

Note: ML-level failures are not addressed in this thesis



- **Go + libp2p** over TCP/IP streams (+ light HTTP per node);
- networks up to **100 nodes**;
- modes: **synchronous / externally-synchronized / asynchronous**;
- **hubs emerge within the first few cycles**
 - matching theory & simulation.



Lift

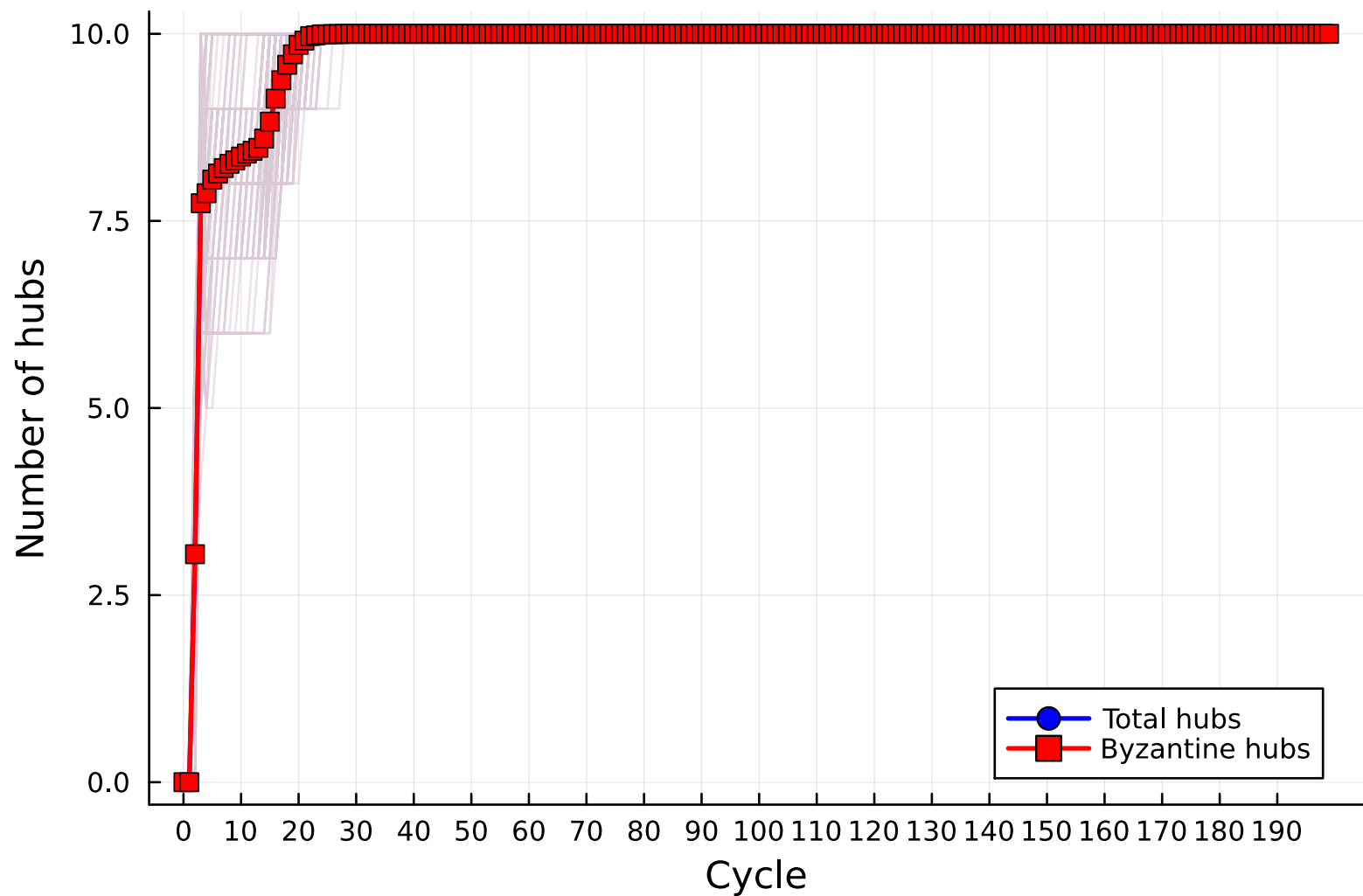
Coordinated attack — lying about neighbors

addresses of all colluding nodes: **colluders**

backward list: **backward_peers**

```
1 loop
2   (request, peer) ← receive()
3   if request == CACHE_REQUEST
4     backward_peers.add(peer)
5     colluders.shuffle()
6     modified_cache ← colluders[0:c]
7     send(modified_cache, peer)
```

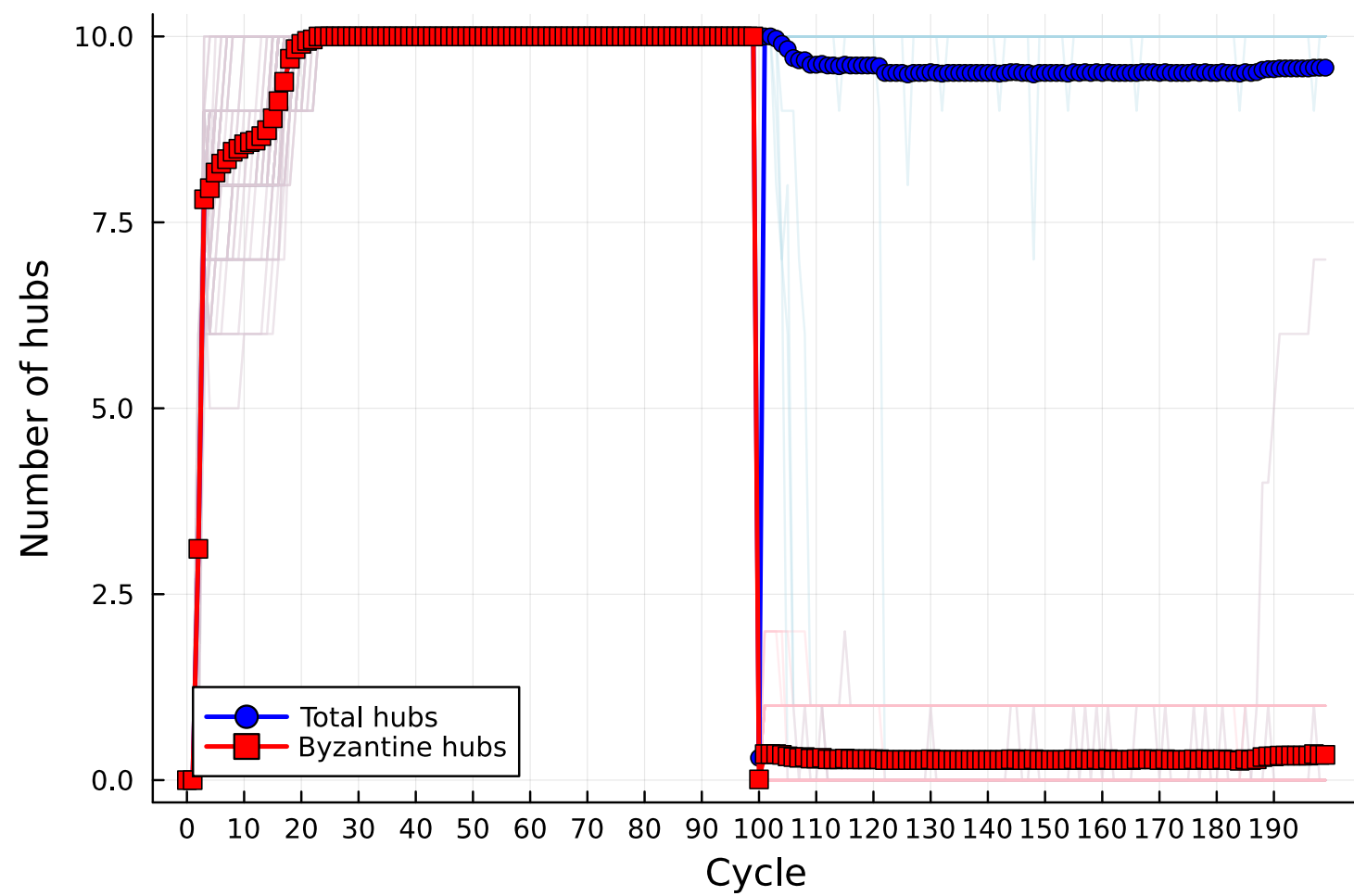
Each colluding node answers with a fake cache that only contains **other colluding nodes**, inflating their visibility.



After convergence, correct nodes **deterministically** re-draw the hubs from a shared random seed — attackers can no longer keep their hub position (effective against infiltration at low collusion rates).

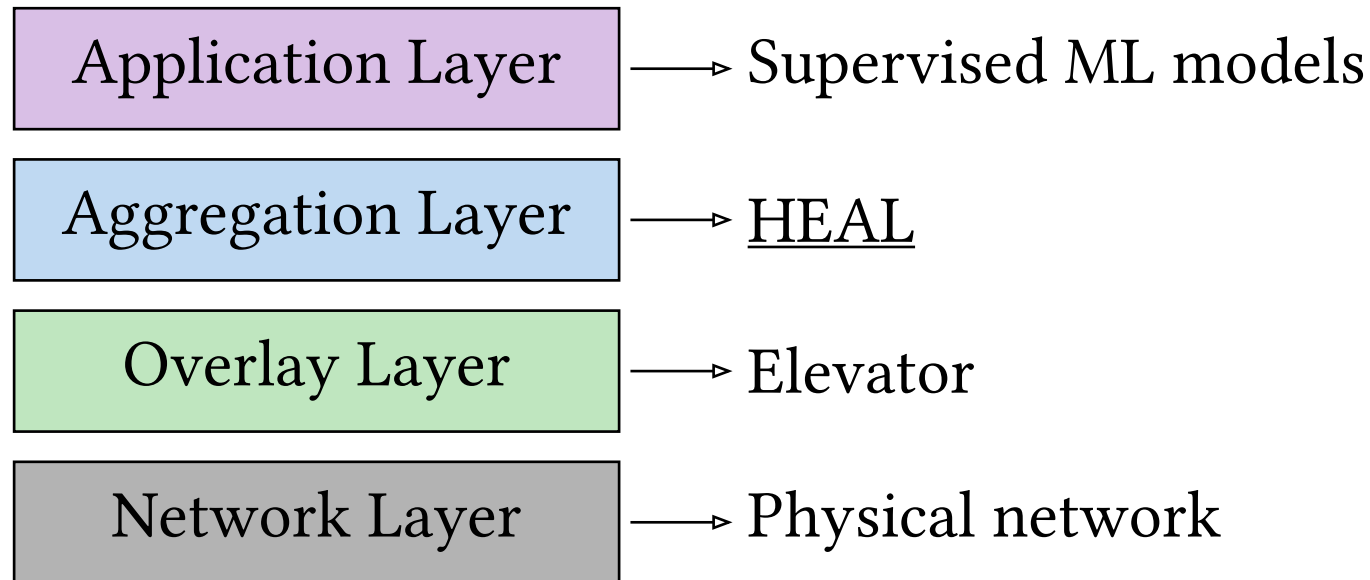
Hub redistribution

```
current hub list: H
network size: N
target hubs: h
1 seed  $\leftarrow$  sort(H)
2 prng  $\leftarrow$  Random(seed)
3 selected  $\leftarrow$  {}
4 while selected.size() < h
5   | id  $\leftarrow$  prng.nextInt(N)
6   | if id not in selected
7   |   | selected  $\leftarrow$  selected  $\cup$  {id}
8 H  $\leftarrow$  selected
```



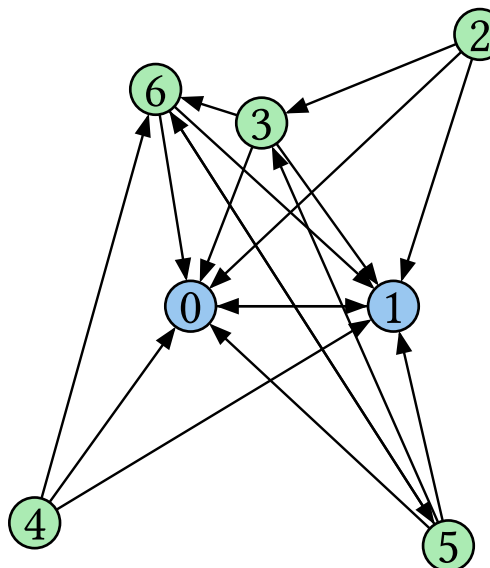
HEAL





1) Each node train a local model

2) Each node sends its model to a (random) hub



3) Hubs receive and aggregate models

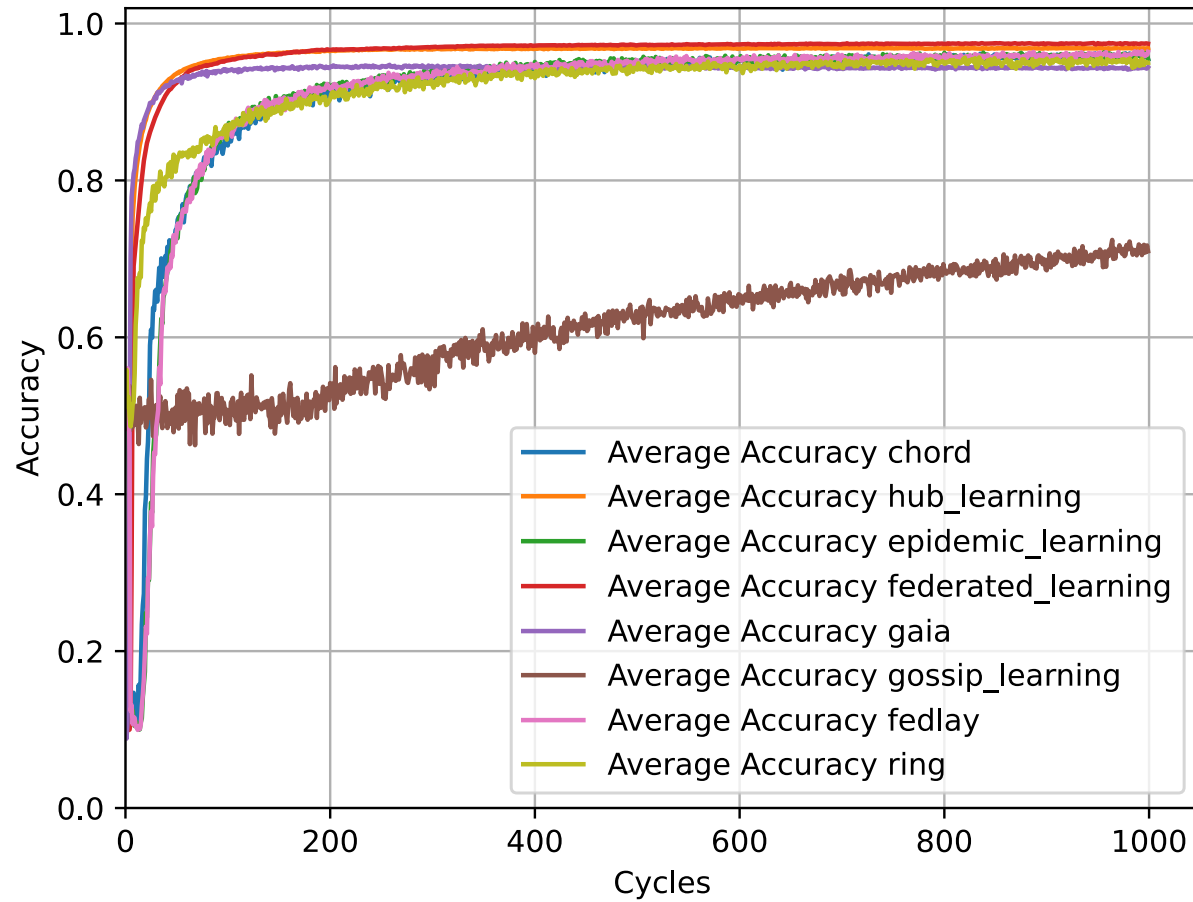
4) Hubs compute a global model together

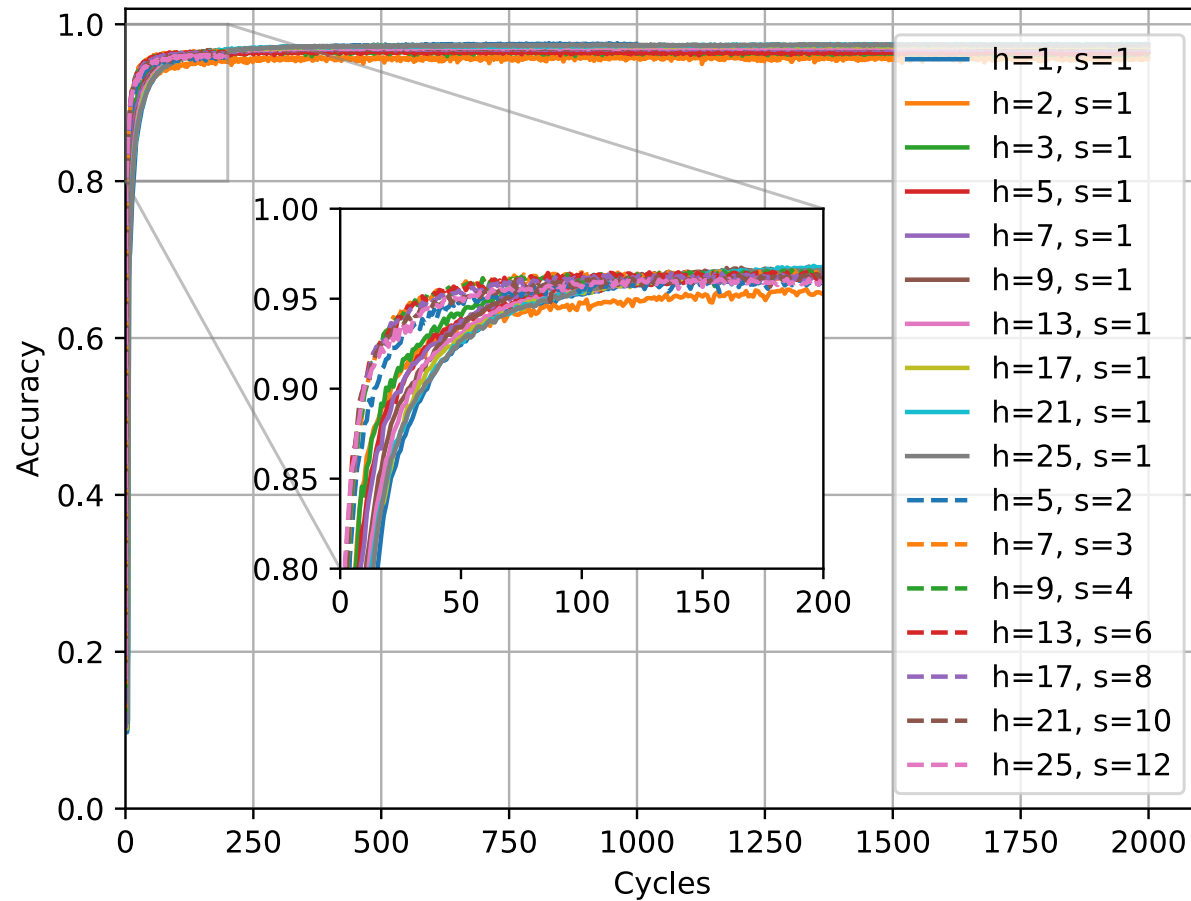
5) Hubs send back the global model to the nodes

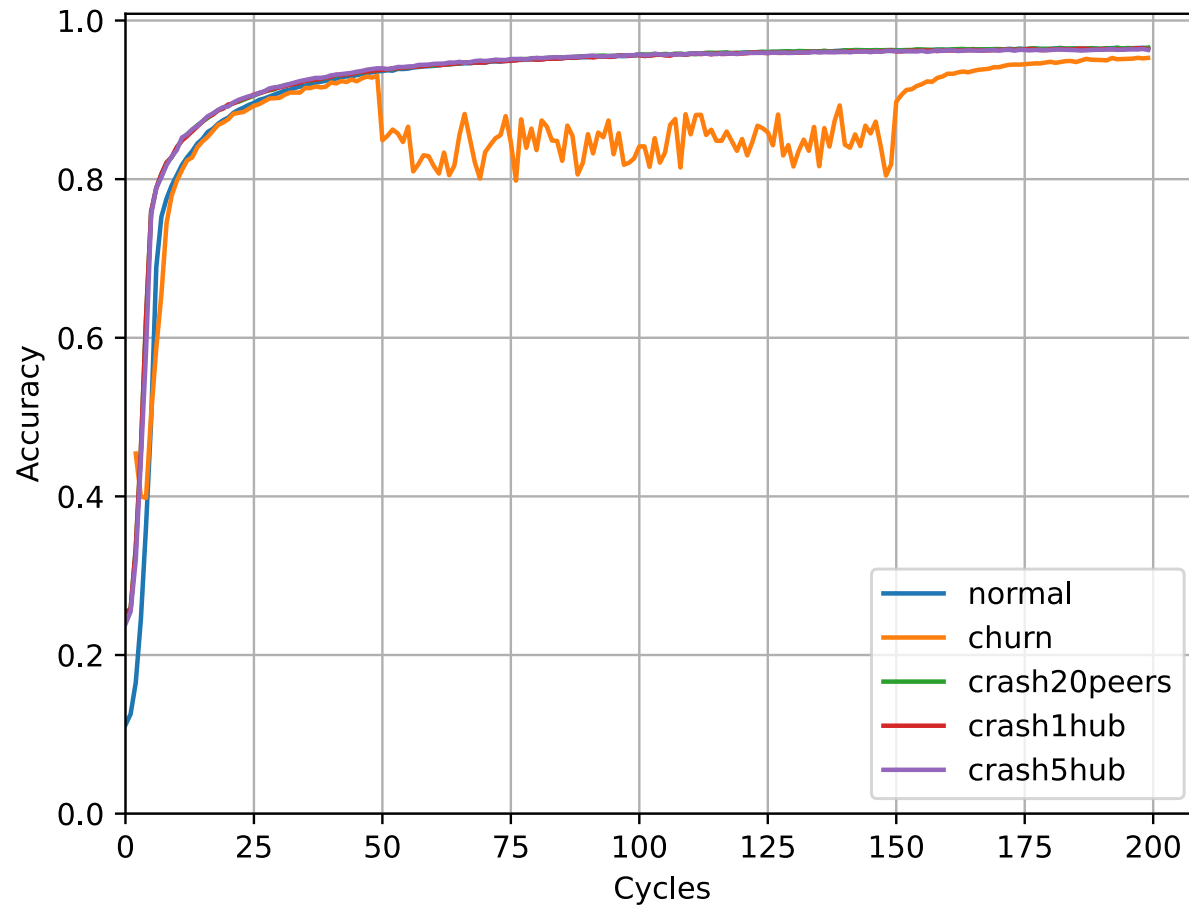
[7] M. A. Legheraba, S. Galkiewicz, M. Potop-Butucaru, and S. Tixeuil, “HEAL: Resilient and Self-* Hub-based Learning,” in *International Conference on Advanced Information Networking and Applications*, 2025, pp. 200–211.

Context of experiments

- **Peer-to-Peer:** On the PeerSim simulator.
 - Each generated network had 100 nodes, and the Hub-based topology had 5 hubs
 - In addition to the crash-free context, we ran the algorithm during a context of crash, a context of churn and a context of an attack on the hubs
- **Decentralized Learning:** On the gossip simulator
 - **ML models:** Logistic Regression & LeNet5
 - **Datasets:** Spambase and MNIST
 - Comparison with Federated Learning, Gossip Learning, Epidemic Learning, GAIA, Chord-based Learning, Fedlay

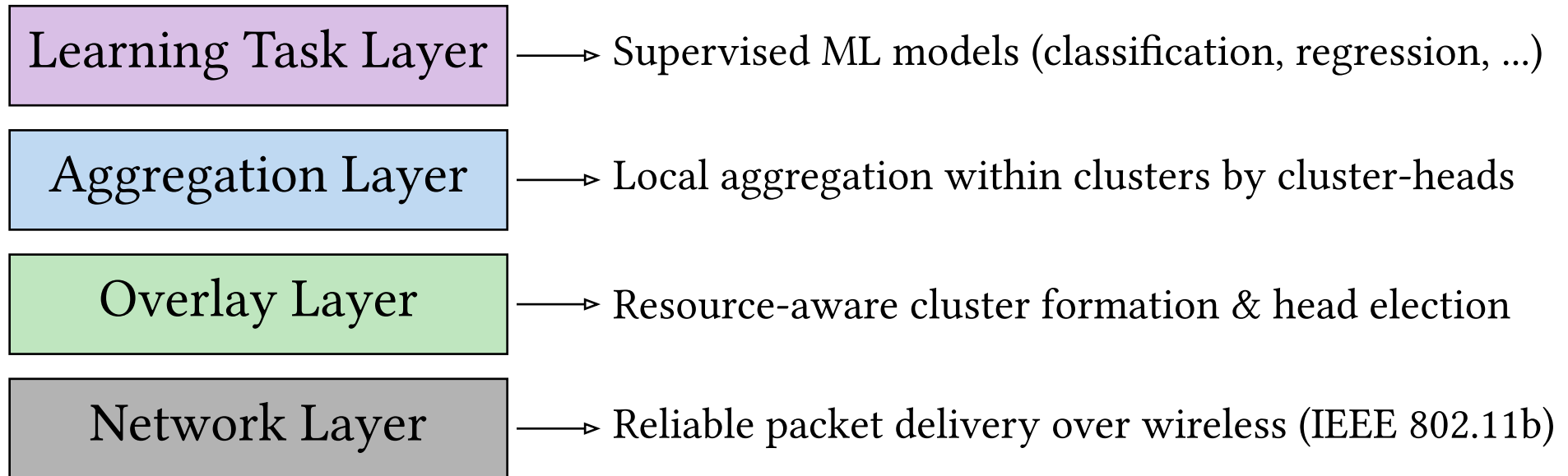






FLAIR

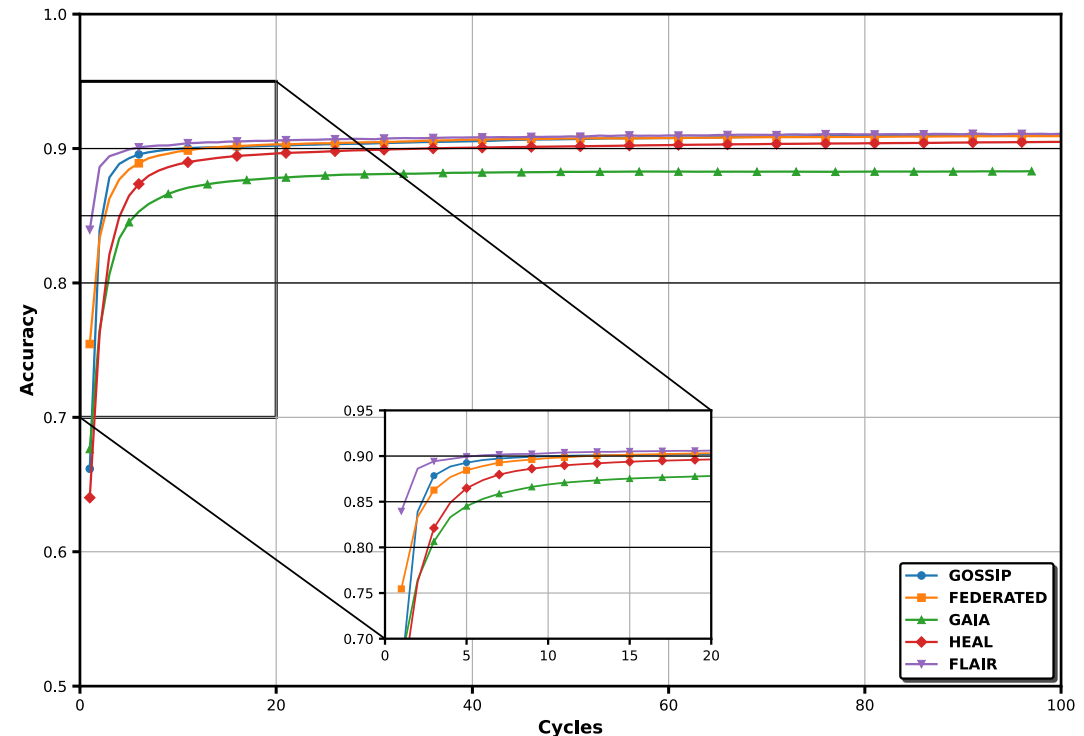




FLAIR operates in **rounds** (LEACH-style clustering):

1. Each node computes its **resource score** (CPU, RAM, GPU, bandwidth).
2. Each round, a fraction of nodes **elect themselves as cluster-heads** (CHs)
— the role rotates.
3. Nodes send their **local models** to their nearest CH, which aggregates them.
4. CHs **change every round** → the aggregated models keep **mixing across clusters**.

- Static network (100 nodes): **highest accuracy** ≈ 0.91 (C-FL / HEAL / Gossip ≈ 0.90 , Gaia ≈ 0.88)
- Convergence up to **2.5x faster** than Gaia
- Resilient to **permanent / temporary / random crashes** (up to 90% nodes)
- Tested under **mobility** and in a **smart farming** scenario



Conclusion



	Decentralized	Fast convergence	Fault-tolerant	Resistant to colluding	Learning	Network type
Elevator	✓	✓	✓	✗	—	P2P
Elevator + Lift	✓	✓	✓	✓	—	P2P
Elevator + HEAL	✓	✓	✓	✗	✓	P2P
FLAIR	✓	✓	✓	✗	✓	Wireless

- **Simple ML models.** Benchmarks are modest — logistic regression on Spambase, LeNet5 on MNIST. No large-scale task, so convergence and accuracy conclusions remain to be confirmed on more ambitious models.
- **Combinatorial explosion.** Network parameters (N, c, h), failure/churn scenarios, fraction of malicious nodes, ML hyper-parameters, and seeds define an enormous space — only a small fraction can be explored.

- **Capability-aware hub election** (heterogeneous nodes) and sensitivity to the initial topology.
- Optimise Elevator (message complexity, memory, simplicity).
- Address **data heterogeneity** in HEAL (personalisation, locally-weighted aggregation).
- A unified **Python simulator** for networking + ML.

This work has already begun during a **3-month internship at NII (Tokyo)**.

- Better robustness at the Overlay layer
- Byzantine robustness in the learning layer (poisoning & privacy attacks)
- Extend HEAL to unsupervised and reinforcement learning
- Vertical federated learning (different features per participant).
- Formal convergence guarantees for HEAL.

- [8] M. A. Legheraba, M. Potop-Butucaru, and S. Tixeul, “Brief Announcement: A Self-* and Persistent Hub Sampling Service,” in *International Symposium on Stabilizing, Safety, and Security of Distributed Systems*, 2024, pp. 461–465.
- [4] M. A. Legheraba, M. Potop-Butucaru, S. Tixeul, and S. Fdida, “Emergent Peer-to-Peer Multi-Hub Topology,” in *2024 22nd International Symposium on Network Computing and Applications (NCA)*, 2024, pp. 219–226.
- [9] M. A. Legheraba, N. Rachdi, M. Potop-Butucaru, and S. Tixeul, “LIFT: Byzantine Resilient Hub-Sampling,” in *2025 Thirteenth International Symposium on Computing and Networking (CANDAR)*, 2025, pp. 1–9. — **Outstanding Paper Award**
- [7] M. A. Legheraba, S. Galkiewicz, M. Potop-Butucaru, and S. Tixeul, “HEAL: Resilient and Self-* Hub-based Learning,” in *International Conference on Advanced Information Networking and Applications*, 2025, pp. 200–211.
- [10] I. Boutebicha, B. Zaghdoudi, M. A. Legheraba, and M. P. Butucaru, “FLAIR: Distributed Federated Learning with Dynamic Clustering,” in *Proceedings of the 14th International Conference on Networked Systems (NETYS 2026)*, in Lecture Notes in Computer Science. Agadir, Morocco: Springer, 2026.
- [11] M. A. Legheraba, M. Potop-Butucaru, S. Tixeul, and S. Fdida, “Des noeuds anonymes hissés au rang de stars,” in *ALGOTEL 2025–27èmes Rencontres Francophones sur les Aspects Algorithmiques des Télécommunications*, 2025.
- [12] M. A. Legheraba, S. Galkiewicz, M. Potop-Butucaru, and S. Tixeul, “Les étoiles montantes de l'apprentissage distribué,” in *ALGOTEL 2025–27èmes Rencontres Francophones sur les Aspects Algorithmiques des Télécommunications*, 2025.
- A journal extension (journal version of this work) has been **submitted to the IEEE/ACM Transactions on Networking**; we are awaiting the **final review** (currently in **minor revision**).

Bibliography

- [1] B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. y Arcas, “Communication-efficient learning of deep networks from decentralized data,” in *Artificial intelligence and statistics*, 2017, pp. 1273–1282.
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